

DRIMS Administration

Disaster Risk Information Management System

User Manual for Citizens, LGU Officers, and Super-Admins

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1. Introduction and Overview

1.1 System Purpose and Objectives

The Disaster Risk Information Management System (DRIMS) is a web-based application designed to facilitate the collection, assessment, and management of public complaints and disaster incident reports. The system leverages Artificial Intelligence (AI), specifically Natural Language Processing (NLP) and Adaptive DBSCAN algorithms, to assist governing bodies in categorizing reports, identifying high-risk geographic clusters, and expediting response efforts.

DRIMS aims to reduce manual processing overhead for Local Government Units (LGUs) while providing citizens with a transparent and accessible platform for reporting hazards. By combining automated data classification with human oversight, the system ensures that critical incidents are prioritized accurately and that response resources are deployed efficiently across affected areas.

1.2 Understanding User Roles

DRIMS enforces a strict three-tier role hierarchy to control data access and system modification. Each role has distinct responsibilities and permissions that govern how users interact with the platform.

- **Citizen:** The public-facing role. Citizens are permitted to submit incident reports, attach supporting evidence, track the resolution status of their submissions, and view aggregated data on a public incident map.
- **LGU (Local Government Unit):** The operational command center role. LGU officers are responsible for reviewing incoming citizen reports. They interact with the Human-in-the-Loop (HITL) AI verification process to validate algorithmic assessments, manage report statuses, and utilize the Strategic Heatmap for spatial data analysis.
- **Super-Admin:** The technical oversight role. Super-Admins control global system parameters, manage user role assignments, and maintain the NLP Dictionary which dictates how the AI interprets and categorizes text data.

2. Getting Started

2.1 Account Registration and Authentication

Access to DRIMS requires a registered account. Follow the steps below to create a new user profile and authenticate into the system.

To register a new account:

1. Navigate to the DRIMS login page in your web browser.
2. Locate the **Register** button in the center-right of the authentication panel and click it.

3. In the registration form, input your personal details into the provided text fields.
4. Click the **Submit** button at the bottom of the form.
5. Wait for the system to display the *Registration Successful* message.

To log into the system:

1. Navigate to the DRIMS login page.
2. Input your credentials into the **Email** and **Password** text fields located in the center of the screen.
3. Click the **Login** button.
4. Upon successful authentication, the system will route you to your role-specific dashboard.

2.2 Navigating the Main Interface

The DRIMS interface is standardized across all roles utilizing a primary sidebar navigation structure. Understanding the layout will help you move efficiently between modules.

1. Locate the **Main Sidebar** docked permanently on the left-hand side of the screen.
2. Click any of the listed **Module Tabs** (e.g., **Dashboard**, **Reports**, **Map**) within the sidebar to navigate to that specific module. The main content area on the right side of the screen will refresh to display the selected module's interface.
3. To sign out of the system, locate the **Logout** button positioned at the bottom of the **Main Sidebar** and click it.

3. Citizen Portal

3.1 Citizen Dashboard Overview

The Citizen Dashboard serves as the default landing page for Citizen accounts. It displays a quantitative summary of the user's submitted reports and provides direct access to the reporting interface. The central area of the dashboard lists recent activity, while a summarized public incident map occupies the upper-right section of the main content area.

The dashboard is organized into three primary zones: a top statistics bar showing total reports, pending items, and resolved items; a central activity feed listing the most recent submissions; and an embedded public map illustrating nearby incidents submitted by other citizens in your area.

3.2 Submitting a New Incident Report

Citizens must use the designated form to submit incident data to the LGU. Ensure all required fields are completed before finalizing the submission.

1. On the **Main Sidebar** located on the left-hand side of the screen, click the **Submit Report** tab.
2. In the main content area, locate the **Incident Title** text field at the top of the form and input a brief descriptive title.
3. Locate the **Description** text box below the title and input the full details of the complaint or incident.
4. Locate the **Upload Evidence** button in the lower section of the form and click it to attach supporting files.
5. Select the relevant file from your local device dialog and confirm the upload.
6. Click the **Submit** button located at the bottom right corner of the form.
7. Verify that the system displays the *Report Submitted Successfully* confirmation message.

3.3 Tracking the Status of Your Reports

Citizens can monitor the progression of their submitted reports as they are processed by the LGU. The status column provides real-time updates on each report's lifecycle.

1. On the **Main Sidebar**, click the **My Reports** tab.
2. In the main content area, locate the **Reports Data Table**.
3. Review the **Status** column within the table to determine the current state of a specific report. System statuses will display as *Pending*, *Verified*, or *Resolved*.

4. LGU Command Center

4.1 LGU Dashboard and Analytics Overview

The LGU Dashboard provides a statistical overview of system-wide incident data. The main content area contains aggregated analytics charts detailing report volumes, categorization metrics, and average resolution times. This data assists LGU officers in assessing overall operational performance.

Key metrics displayed include the total number of incoming reports for the current day, the breakdown by incident category (e.g., Flooding, Fire, Landslide), and the average time taken from submission to verification. The dashboard serves as a central monitoring hub for managing the LGU's disaster response workload.

4.2 Navigating the Strategic Heatmap

The Strategic Heatmap utilizes Adaptive DBSCAN to visualize spatial concentrations of incidents. This tool helps officers identify high-risk zones and allocate resources accordingly.

To toggle between clusters and individual markers:

1. On the **Main Sidebar**, click the **Heatmap** tab.
2. Locate the **Map Controls Panel** overlay positioned in the top-right corner of the map interface.
3. Click the **Clusters** toggle button to group neighboring incidents into color-coded density zones.
4. Click the **Markers** toggle button to ungroup the data and display individual incident pinpoints.

To apply geographic and variable filters:

1. Locate the **Filter Panel** positioned directly above the map interface.
2. Click the **Barangay** dropdown menu and select a specific jurisdiction to isolate the map view to that boundary.
3. Click the **Status** dropdown menu to filter visible data by current system state (e.g., *Pending, Verified*).
4. Click the **Priority** dropdown menu to filter visible data by severity level.
5. The map will automatically refresh to reflect the selected filter parameters.

4.3 Incident Complaint Management

LGU officers must review citizen submissions using the HITL verification workflow. This process ensures that AI classifications are validated by a human operator before actions are taken.

To execute the HITL Verification Process and update statuses:

1. On the **Main Sidebar**, click the **Review Queue** tab.
2. Locate the **Incoming Reports Table** in the main content area.
3. Click the **Review** button located on the right side of a specific report row.
4. A **Verification Panel** will appear on the right side of the screen displaying the raw citizen complaint.
5. Below the complaint text, review the **AI Assessment Data**, which displays the NLP system's suggested category and priority level.
6. To accept the AI's suggestion, click the **Validate** button at the bottom of the **Verification Panel**.
7. To reject or modify the suggestion, click the **Override** button, manually select the correct category from the dropdown menu, and click **Save Changes**.
8. To update the overall state of the report, locate the **Update Status** dropdown menu at the top of the **Verification Panel**.

9. Select the appropriate new status (e.g., *Verified*).
10. Click the **Confirm** button. The system will display a *Status Updated* notification.

5. Super-Admin Operations

5.1 User Management and Role Allocation

Super-Admins hold the authority to modify the access privileges of any account. This function is critical for onboarding new LGU personnel or adjusting permissions as organizational needs change.

1. On the **Main Sidebar**, click the **User Management** tab.
2. Locate the **User Roster Table** in the center of the screen.
3. Locate the row corresponding to the target user account and click the **Edit** button on the far right of that row.
4. In the resulting modal window, locate the **Role** dropdown menu.
5. Select the new target role (e.g., LGU, Citizen).
6. Click the **Save** button at the bottom of the modal window. The system will display a *Role Updated* notification.

5.2 Dictionary Management

The Dictionary Management module allows Super-Admins to calibrate the NLP engine by defining exactly which keywords map to specific incident categories. Maintaining an accurate dictionary is essential for ensuring that citizen reports are classified correctly.

1. On the **Main Sidebar**, click the **Dictionary Configuration** tab.
2. Locate the **Keyword Mapping Table** in the main content area.
3. To add a new keyword, click the **Add Entry** button situated above the top-right corner of the table.
4. In the new entry form, type the target word into the **Keyword** text field.
5. Select the corresponding category from the **Mapped Category** dropdown menu.
6. Click the **Submit Entry** button.
7. To modify an existing keyword, locate its row in the table, click the **Edit** button, adjust the mapped category, and click **Save**.
8. To remove a keyword entirely, click the **Delete** button on the corresponding row and confirm the action in the subsequent prompt.

5.3 Global Configuration Control

Super-Admins manage high-level environmental variables affecting system behavior. These settings impact all users and should be adjusted with care.

1. On the **Main Sidebar**, click the **Global Settings** tab.
2. Locate the **System Parameters Panel** in the center of the screen.
3. Modify numeric values within the provided text fields to adjust backend operational thresholds (e.g., maximum upload sizes, session timeouts).
4. Click the **Apply Configuration** button located at the bottom of the panel. The system will display a *Configuration Applied* notification.

Global Configuration Parameters

Parameter	Default Value	Description
Max Upload Size	10 MB	Maximum file size allowed for evidence attachments
Session Timeout	30 minutes	Duration of inactivity before automatic logout
HITL Queue Limit	100 reports	Maximum reports queued for LGU review before alert
Cluster Radius	500 meters	Radius used by DBSCAN for grouping markers

6. Troubleshooting and Support

6.1 Common Issues

This section addresses frequent problems encountered by DRIMS users and provides direct solutions to resolve them.

- **Authentication Failure:** If the system returns an *Invalid Credentials* error upon login, verify that the email format is correct and the password matches the registered record. Ensure the caps lock key is disengaged.
- **Map Rendering Errors:** If the Strategic Heatmap fails to load geographic boundaries or markers, refresh the browser window. Ensure the active device has a stable internet connection capable of fetching external map tiles.

- **File Upload Rejection:** If the system returns an *Upload Failed* message during citizen reporting, verify that the attached evidence file does not exceed the maximum file size limit defined by the Super-Admin.

6.2 Glossary of Terms

- **Cluster:** A visual grouping on the Strategic Heatmap representing a high spatial density of individual incident reports.
- **DBSCAN (Adaptive):** Density-Based Spatial Clustering of Applications with Noise. The underlying algorithm DRIMS uses to group neighboring data points into Clusters on the heatmap.
- **HITL (Human-in-the-Loop):** A system design pattern where an artificial intelligence makes an assessment or prediction, but a human operator (the LGU officer) must manually review and authorize that decision before it becomes final.
- **Marker:** A specific, individual pinpoint on the Strategic Heatmap representing a single, isolated incident report.
- **NLP (Natural Language Processing):** The artificial intelligence module that reads raw text submitted by citizens and attempts to categorize the complaint based on configured dictionary keywords.